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MRV: INTERPRETING MEDICAL REPORTS INTO VOICE IN REGIONAL LANGUAGES

MINI PROJECT REPORT

Submitted by

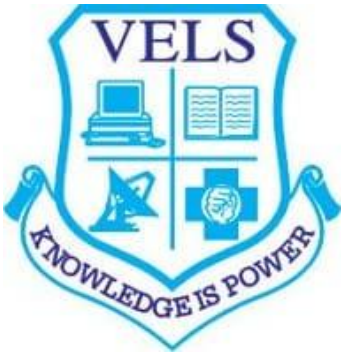
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BONAFIDE CERTIFICATE

Certified that this project report "**MRV: INTERPRETING MEDICAL REPORTS INTO VOICE IN REGIONAL LANGUAGES**", is the Bonafide work of **M. DHARSHINI - 24619111**, **A. DEVI- 24619108**, **B. JOTHILAKSHMI - 24619106** carried out the project under my supervision. Certified further, that to the best of my knowledge the work reported here does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate

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ABBREVIATIONS

ABBREVIATION	FULL FORM
MRV	Medical report voice
AI	Artificial Intelligence
ML	Machine Learning
OCR	Optical Character Recognition
LLM	Large Language Model
TTS	Text-to-Speech
API	Application Programming Interface
PDF	Portable Document Format
CSS	Cascading Style Sheets
HTML	Hyper Text Markup Language
JS	Java Script

ABSTRACT

In Understanding medical reports is difficult because of complex terminologies used in medical reports. This study proposes the creation of a healthcare application using AI for interpreting the medical reports and making the reports more user-friendly. Medical reports such as CBC, thyroid, and Pulse Reports, cardiology reports and general laboratory reports can be uploaded by the users into the app and using Optical Character Recognition (OCR), data in the report can be extracted. The Large Language Model (LLM) uses this information to interpret the medical report in simple language. Explanation of this interpretation is given in the form of both text and audio in regional languages using Text-to-Speech (TTS) Technology. In addition, the app provides pictorial explanation of body parts involved in the medical condition, important terms in the medical field, and tutorial videos for the first-time users. In addition to these features, Report History allows the user to view past reports while Health Trend Analysis allows the user to track health parameters.

Keywords AI, OCR, LLM, TTS, Medical Report Interpretation, Regional Languages, Report History, Health Trend Analysis.

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

Medical reports contain important information about a person’s health. However, they often include complex medical terms and abbreviations and numerical values that are often difficult for ordinary patients to understand. For example, a blood report may contain terms such as hyperlipidemia or hemoglobin, which patients may not fully understand. Another major problem is the language barrier. Even if patients receive their reports, they may struggle to understand the medical information when it is presented in a language, they are not comfortable with. To address these problems, we developed MRV, an AI-based patient-support application. It processes medical reports and generates simpler explanations in the user's preferred regional language. Our system combines text extraction, AI-based simplification, text-to-speech, report history, and health-trend visualization." Users can upload an image of their medical report through the application. OCR technology is used to extract text from the uploaded report. The extracted text is processed using the Qwen (LLM). Gemini simplifies the medical content into easy-to-understand language. The application also supports multiple languages for better accessibility. Text-to-Speech (TTS) allows users to listen to the simplified report. The application stores previous reports for future reference. The Health History feature helps users access their earlier medical reports. The Health Trend feature helps users track changes in their health information over time. Users can compare information from their current and previous reports. This helps them maintain a convenient record of their medical information. Overall, the application makes medical reports easier to understand and manage. It is designed as a supportive tool and does not replace professional medical advice.

Research Gap

Lack of OCR for Scanned Reports: Many existing systems work mainly with digital or text-based medical reports. Many existing applications cannot easily read medical reports uploaded as images or scanned documents.

Limited Regional Language Support: Most existing system explain medical reports only in English. People who understand Tamil or other regional languages may find it difficult.

Limited Patient-Friendly Visualization: Many existing applications show only text-based explanation and the application does not support color coded indicators. Patients may find it difficult to understand whether their medical results are normal or abnormal.

Limited Audio Support: Most existing applications display explanations as text. Some patients may find difficulties in reading the text, the existing system provides limited audio support.

Problem Statement

Traditional medical reports contain clinical information using complex medical terminology, abbreviations, and numerical values. Patients often struggle to understand these reports, especially when they are provided in unfamiliar languages. This creates several significant problems:

1.Complex Medical Terminology: Medical reports contain technical terms and abbreviations that are difficult for ordinary patients to understand without professional assistance.

2.Language Barriers: Reports may be available only in certain languages, making it difficult for patients to understand their medical information in their preferred regional language.

3.Limited Accessibility: Patients who have difficulty reading medical reports may struggle to interpret the information without simple explanations or voice assistance.

These challenges can reduce patients' understanding of their medical conditions and make it harder for them to interpret the information provided in their reports. The core problem is the lack of an integrated, accessible system that can extract medical information, simplify complex terminology, provide regional-language explanations with voice output, and help users review their health trends over time.

Objectives of the Project

The main objectives of this project are: Develop an AI-based application that explains medical reports in simple, patient-friendly language.

- Extract report text from images and supported PDF files using OCR/document processing.
- Use a Qwen LLM to simplify medical terminology while retaining key information.

- Present explanations in the user-selected regional language and provide text-to-speech output.
- To display the reports using color-coded indicators and health-trend visualization.
- To allow the users to save and review their previous medical reports.

Scope of the Project

The scope of MRV (Medical Report Voice) is to develop an AI-powered application that helps patients understand their medical reports in simple and understandable language. The system allows users to upload medical reports such as blood test reports, general laboratory reports and cardiology reports in PDF and image formats, extracts text using PDF.js and Tesseract.js OCR, and uses the Qwen Large Language Model to generate patient-friendly explanations. It supports multiple languages, The language selection screen includes English, Tamil, Telugu, Hindi, Malayalam, Kannada, Marathi, Bengali, Gujarati, Punjabi, Japanese, Chinese, Korean, and Thai. Users can select their preferred language for the report explanation and voice output. enabling users to understand their medical information in their preferred regional language, and provides text-to-speech functionality for audio-based explanations. The application also displays colour-coded health risk indicators and trend graphs to help users monitor changes in medical values over multiple visits. Additionally, it provides user authentication and report history features to support secure access and easy retrieval of previous reports. MRV aims to improve healthcare accessibility and patient awareness, the application is intended for patients, elderly users, families who find medical terminology or language a barrier but it is designed to support understanding of medical reports and does not replace professional medical advice or diagnosis.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE OVERVIEW

Existing research explores the use of Artificial Intelligence (AI), Optical Character Recognition (OCR), and Natural Language Processing (NLP) to simplify complex medical reports. Studies using ChatGPT and other large language models have shown the potential to generate patient-friendly explanations of medical information. Researchers have also explored translating medical reports into regional languages to improve patient understanding. Researchers have also explored visual methods for presenting medical information. Graphs can help users compare medical values across multiple visits and observe changes over time. Colour-coded indicators can draw attention to values that fall within predefined normal, moderate, or high-risk categories. However, these indicators must be based on appropriate medical reference ranges and should not be treated as a final diagnosis. Combining text explanations with visual representations can make medical information easier to follow. Despite these developments, several challenges remain, including inaccurate text extraction, difficulty in preserving medical meaning, inconsistent regional-language output, limitations in voice support, and the reliability of AI-generated explanations. Many existing approaches focus mainly on simplifying reports or evaluating readability, while other useful functions, such as voice-based access and tracking medical values over multiple visits, may require separate tools.

Based on these research developments, the MRV project integrates OCR-based text extraction, Qwen Large Language Model-based medical report simplification, regional-language support, and Text-to-Speech functionality. The system also provides colour-coded indicators, trend graphs, and report history to help users understand their medical information and compare selected values across visits. By combining these features in a single application, MRV aims to make medical reports more accessible and easier for patients to understand. The application is designed to support patient awareness and understanding and does not replace professional medical advice or diagnosis.

2.2 LITERATURE SURVEY

1.Sarangi et al. (2023) – Assessing ChatGPT's Proficiency in Simplifying Radiological Reports for Healthcare Professionals and Patients.

Sarangi et al. [3] evaluated ChatGPT-3.5 for simplifying radiological reports and supporting medical communication. The researchers selected nine radiology reports from different imaging modalities and asked ChatGPT to explain procedures, summarize important findings, simplify technical language, and translate information into Hindi. Eight radiologists assessed the generated responses for detail, accuracy, simplification, grammar, and translation quality. The results indicated that ChatGPT could simplify technical information while preserving important findings in many cases. However, the Hindi translations were not considered suitable for patient communication, demonstrating that language generation quality may vary across languages. This finding is particularly relevant to MRV because the project aims to provide explanations in regional languages. It highlights the need to evaluate language consistency and ensure that translated explanations remain clear and accurate.

2.Park et al. (2024) – Patient-Centered Radiology Reports with Generative Artificial Intelligence: Adding Value to Radiology Reporting

Park et al. [5] explored the use of generative artificial intelligence to improve the patient-friendliness of radiology reports. Their research examined AI-generated summaries, explanations, and recommendations, with an emphasis on report quality and accuracy. The study investigated whether generative AI could make technical radiological information more accessible without losing important clinical meaning. Its approach demonstrates the potential of AI-assisted report generation to support patient-centred communication. However, the quality and consistency of generated information remain important concerns when AI is used in healthcare. This research is relevant to MRV because the project uses Qwen to simplify medical information and aims to present explanations that ordinary users can understand. It also supports the need to assess generated content carefully before patients rely on it.

3.Sterling et al. (2024) – Patient-Readable Radiology Report Summaries Generated via Large Language Model: Safety and Quality

Sterling et al. [6] investigated the safety and quality of patient-readable radiology summaries generated using a large language model. Eight board-certified physician reviewers assessed 1,982 AI-generated summaries derived from CT, MRI, ultrasound, and X-ray reports. The

summaries were evaluated against the corresponding original reports using predefined safety and quality criteria. This approach provided a broader assessment of AI-generated medical explanations across different imaging modalities. The study emphasizes that a report may be easy to read but still contain an omission or an incorrect statement, making factual accuracy and completeness essential evaluation criteria. This research is directly relevant to MRV because the system generates patient-friendly explanations from extracted medical text. It indicates that MRV should preserve the original findings and undergo appropriate testing for accuracy and safety.

4.Schmidt et al. (2024) – Simplifying Radiologic Reports with Natural Language Processing: A Novel Approach Using ChatGPT in Enhancing Patient Understanding of MRI Results

Schmidt et al. [4] studied the use of ChatGPT to simplify MRI reports of the knee. The researchers generated simplified versions of reports with different levels of complexity and evaluated them with medical professionals and patients without medical backgrounds. The evaluation considered factual correctness, completeness, readability, and the possibility of harmful conclusions. The findings indicated that simplified reports could improve patient understanding of the main findings and make complex terminology easier to interpret. Nevertheless, the authors emphasized that AI-generated explanations cannot replace direct communication between patients and healthcare professionals. This study is relevant to MRV because it demonstrates the importance of adapting technical medical content to the patient's level of understanding. MRV extends this concept by combining AI-based simplification with regional-language and text-to-speech features.

5.Gupta et al. (2025) – Provision of Radiology Reports Simplified with Large Language Models to Patients with Cancer: Impact on Patient Satisfaction

Gupta et al. [8] investigated the use of large language models to simplify radiology reports for patients with cancer. In the first phase, the researchers compared five models, including GPT-4o, Google Gemini, Claude Opus, Llama 3.1, and Phi-3.5-mini, using different prompts to simplify oncology CT report impressions. The generated reports were assessed using readability measures, and selected outputs underwent qualitative evaluation. In the second phase, 100 oncology patients were assigned to groups receiving either the original report impression or a simplified version generated under radiologist supervision. The study found that patients receiving simplified reports demonstrated better understanding and confidence in interpreting their medical information. However, the study also showed the importance of

human oversight, as some generated reports required correction. This research is especially relevant to MRV because it evaluates Google Gemini alongside other language models and supports the development of AI-based medical report simplification systems.

6.Sun et al. (2026) – Patient-Friendly Simplification and Translation of Neuroradiology Impressions Using Artificial Intelligence

Sun et al. [6] investigated the use of artificial intelligence to simplify and translate neuroradiology reports. The researchers collected 500 reports covering brain MRI, head CT, and spine MRI. They used GPT-4o to simplify the reports to approximately a middle-school reading level and translated a subset of the simplified impressions into Spanish. The outputs were assessed by neuroradiologists for clinical accuracy and by qualified medical interpreters for translation quality. The results showed improved readability, low rates of clinically relevant errors, and translation quality comparable to manual translations in the evaluation performed. However, the study focused on neuroradiology reports and Spanish translation, so the findings cannot automatically be generalized to all medical report types and regional languages. This research is highly relevant to MRV because it demonstrates how AI-based simplification and translation can work together to make medical information easier to understand.

CHAPTER 3

SYSTEM DESIGN

3.1 EXISTING SYSTEM

Overview

Existing medical report systems mainly present test results, medical terminology, abbreviations, and numerical values in a technical format. Patients often depend on doctors or healthcare professionals to understand the meaning of these reports. Some existing applications use Optical Character Recognition (OCR) to extract text from medical documents, while others use Artificial Intelligence (AI) and Natural Language Processing (NLP) to generate simplified explanations. Certain systems also provide translation features to improve accessibility. However, these features are often available separately, and patients may find it difficult to understand their reports when language support, voice assistance, and medical history tracking are limited.

Limitations of the Existing system

1. Complex Medical Terminology: Medical reports contain technical terms that are difficult for ordinary patients to understand.
2. Limited Language Support: Some existing systems provide medical report explanations only in a limited number of languages. This creates difficulties for patients who prefer regional languages, as they may not fully understand the information provided in English or other unfamiliar languages.
3. Inaccurate Text Extraction: OCR tools may extract incorrect or incomplete text from blurry, handwritten, or low-quality medical reports.
4. Limited voice explanation: The system has limited ability to explain medical reports through spoken audio, which may make it difficult for users who prefer listening to reading
5. Difficulty Tracking Medical History: Comparing previous and current medical reports may require considerable manual effort. Patients may find it difficult to identify changes in medical values over time, making it harder to monitor health trends across multiple visits.

3.2 PROPOSED SYSTEM

The proposed MRV (Medical Report Voice) system is an AI-based application designed to help patients understand their medical reports easily. It allows users to register, log in, and upload medical reports in PDF, JPG, and PNG formats. PDF.js extracts text from digital PDFs, while Tesseract.js OCR recognizes text from scanned reports and images. The extracted text is processed using Qwen Large Language Model (LLM) to generate simple, patient-friendly explanations. The system provides explanations in the user's selected regional language and uses Text-to-Speech (TTS) to convert them into spoken audio. It also displays colour-coded medical indicators, categorizing selected results as Normal, Moderate, or High based on predefined rules. Health-trend graphs help users compare medical values across multiple visits, while report history allows them to access previously processed reports.

Overall, MRV combines AI-based simplification, regional-language support, voice output, visual indicators, trend graphs, and report history in a single application. It aims to improve the accessibility and understanding of medical information. However, it is designed to support patient understanding and does not replace professional medical advice or diagnosis.

3.3 SYSTEM ARCHITECTURE

Architectural diagram:

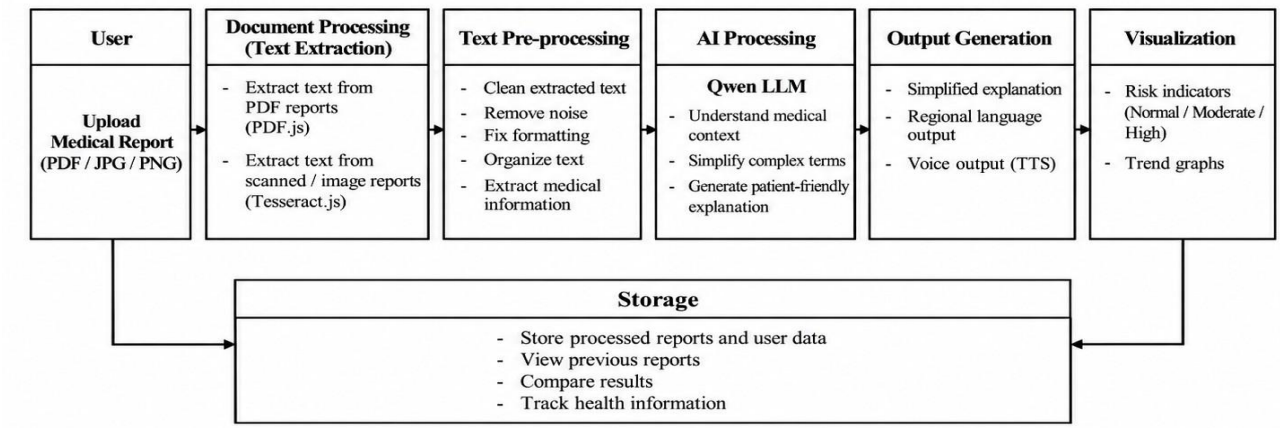


Figure3.1 Proposed system architecture

Data Flow Sequence:

User Login → Report Upload → Text Extraction → Text Preprocessing → AI Simplification → Language Processing → Voice and Visual Output → Report History

This workflow illustrates how information moves through MRV, from the initial report upload to the presentation and storage of the results.

3.4 SOFTWARE REQUIREMENTS

Table 3.4.1 Software Tools

Software	Description
Operating System	Windows 10/11 or Ubuntu 20.04 or later
Programming Languages	HTML5, CSS3, JavaScript
Development Environment	Visual Studio Code or another suitable code editor

Table 3.4.2 Libraries and Components

Component	Specification / Purpose
PDF.js	Extracts text from text-based PDF medical reports
Tesseract.js OCR	Extracts text from scanned medical reports and images
Qwen LLM	Simplifies medical terminology into patient-friendly explanations
Text-to-Speech (TTS)	Reads the simplified explanation aloud in the selected language
Charting Library	Displays trend graphs for selected medical values, if used in the implementation

Table 3.4.3 Frontend

Software	Description
Web Browser	Google Chrome, Firefox, or Microsoft Edge
Web Technologies	HTML5, CSS3, and JavaScript for the user interface

Table 3.4.4 Data Management

Component	Description
Report Storage	Stores processed report information and user/report history; use the database or storage mechanism implemented in the project
Report History	Supports access to previously processed reports and comparison of selected medical values, where available

CHAPTER 4
IMPLEMENTATION

4.1 METHODOLOGY

Proposed MRV system will follow a systematic method to translate complicated content of the medical report into simpler terms along with an explanation through voice. Proposed MRV system is developed as a web application. User will login and select his preferred language. Then, he will upload a medical report in PDF or image format. As per the format of the document used, text will be extracted from PDF or OCR technique will be applied to extract text from image/scanned medical report. Extracted text will be provided to AI based simplification engine to simplify medical terminologies and contents in simple words. This explanation will be displayed in a box or will be read out by text-to-speech functionality of the browser. MRV system can provide an insight about the overall health risks and save the processed medical report in the local history.

DATASET DESCRIPTION

The proposed system utilizes a pre-trained AI model for simplifying medical reports. Hence, no separate dataset was used for training the model. Instead, a collection of sample medical reports was used to test and evaluate the application's functionality. These sample reports were used to verify the report upload process, text extraction, AI-based simplification, and voice explanation features.

The sample medical reports used for testing are provided through the following links:

* https://github.com/abachaa/MedQuAD?utm_source=chatgpt.com

* https://huggingface.co/datasets/Sadou/medical-reports-simplification-dataset

* https://huggingface.co/datasets/Akhil-reddy/Medical-Balanced?utm_source=chatgpt.com

*https://huggingface.co/datasets/ClinOCR-Bench/ClinOCR-Bench?utm_source=chatgpt.com

*https://github.com/nepython/MedSiML?utm_source=chatgpt.com

Step 1 – User Log-In: The user logs in using the provided log-in portal.

Step 2 – Choice of Language: The user chooses a language that the explanation would be provided in.

Step 3 – Report Upload: The user uploads the medical report in PDF or image form.

Step 4 – Extraction: For extraction of text from PDF documents, we use PDF.js, and for extraction of text from images, we use Tesseract.js.

Step 5 – AI Explanation: The data obtained from the medical report is used to generate an easy explanation.

Step 6 – Voice Explanation: The user uses browser speech synthesis API to hear the simplified explanation in chosen language.

Step 7 – Risk and Report History: The user gets a basic idea about the risk and can access previous reports' history.

4.2 IMPLEMENTATION OVERVIEW

The implementation phase of the MRV (Medical Report Voice) project focuses on developing a web-based system that converts complex medical reports into simple and understandable explanations. The system accepts medical reports in PDF or image format, extracts the report content using text extraction and Optical Character Recognition (OCR), and presents simplified information in the selected language. It also provides voice output to make the explanation accessible to users who find it difficult to read medical terms.

The application is developed using HTML, CSS and JavaScript for the user interface, with Python Flask used for backend processing. PDF.js is used to extract text from PDF documents, while Tesseract.js is used to recognize text from scanned reports and images. An AI-based language processing component is used to simplify the extracted medical information.

4.3 IMPLEMENTATION MODULE

4.3.1 Application Tutorial Video Module

The Application Tutorial Video Module is the first page displayed when the MRV application is opened. It provides an introductory video that explains how to use the application and guides users through its major functionalities.

1.Frontend Technologies: HTML5 is used to embed the video player, CSS3 is used for page layout and styling, and JavaScript handles video-related interactions.

2.Backend Technology: Python Flask is used to serve the application page and its resources.

3.Tools Used: HTML5 Video Player, HTML, CSS and JavaScript.

4.Working Procedure:

When the user opens the MRV website, the introductory page appears with a tutorial video. The video demonstrates the complete application workflow, including login, language selection, medical report upload, AI-based simplification and voice explanation. Users can watch the video to understand the application before proceeding to the login page.

4.3.2 Login and User Interface Module

The Login and User Interface Module provides access to the MRV application. It allows users to enter the application through a simple and user-friendly login interface.

1.Frontend Technologies: HTML5 is used to create the login form and input fields. CSS3 is used for page design and styling, while JavaScript handles form validation and user interactions.

2.Backend Technology: Python Flask manages login requests and server-side operations.

3.Tools Used: HTML5, CSS3, JavaScript and Python Flask.

4.Working Procedure:

After watching the tutorial video, the user is redirected to the login page. The user enters the required login details and submits the form. JavaScript checks whether the required fields are filled and sends the login request to the Flask backend. The backend processes the request according to the configured login logic and returns the response. After successful login, the user proceeds to the next page of the application.

4.3.3 Language Selection Module

The Language Selection Module allows users to select their preferred language for understanding medical reports. It helps make the application accessible to users from different language backgrounds.

1. Frontend Technologies: HTML5 is used to create language selection options, CSS3 provides styling, and JavaScript handles the selected language.

2.Backend Technology: Python Flask receives the selected language information and coordinates the subsequent processing workflow.

3.Tools Used: JavaScript and browser language support.

4.Working Procedure:

After successful login, the application displays the available language options. The user can select English, Tamil, Hindi, Telugu, Malayalam, Kannada, French or Japanese. JavaScript captures the selected language and maintains it for the next stages. The selected language is

passed along with the medical report information to the simplification process. It is also used to select a suitable voice during the voice explanation stage, depending on browser and device support.

4.3.4 Medical Report Upload Module

The Medical Report Upload Module allows users to upload their medical reports in PDF or image format for further processing.

1.Frontend Technologies: HTML5 provides the file upload interface, CSS3 styles the upload section, and JavaScript handles file selection and validation.

2.Backend Technology: Python Flask manages server-side requests and coordinates the report-processing workflow.

3.Tools Used: HTML File Input, JavaScript, PDF.js and Tesseract.js.

4.Working Procedure:

The user selects a medical report from their device using the upload option. The application identifies the uploaded file format and directs it to the appropriate processing method. PDF files are passed to the PDF text extraction process, while image files are passed to the OCR process. The extracted information is then forwarded to the next module for simplification.

4.3.5 Text Extraction and OCR Module

The Text Extraction and OCR Module extracts readable text from uploaded medical reports. It supports both digital PDF documents and scanned image reports.

1.Frontend Technologies: HTML5 displays the extracted information, CSS3 provides formatting, and JavaScript controls the extraction process.

2. Backend Technology: Python Flask coordinates the processing workflow and handles the extracted information.

3.Tools Used: PDF.js and Tesseract.js.

4.Working Procedure:

When the user uploads a PDF document, PDF.js reads the PDF file and extracts its available text content. If the user uploads a scanned report or image, Tesseract.js performs Optical Character Recognition (OCR) to identify and convert the characters into machine-readable text. The extracted text from either method is collected and forwarded to the AI-based simplification module.

4.3.6 AI-Based Medical Text Simplification Module

The AI-Based Medical Text Simplification Module converts complex medical information into simple and understandable language. It helps users understand medical terms, test results and report findings.

1.Frontend Technologies: HTML5 is used to display the simplified explanation, CSS3 provides readable formatting, and JavaScript sends the extracted text and selected language to the backend.

2.Backend Technology: Python Flask receives the extracted medical text, coordinates the simplification process and returns the response to the frontend.

3.Tools Used: AI/LLM-based text simplification component.

4.Working Procedure:

The extracted medical report text and selected language are sent to the Flask backend. The backend forwards the information to the configured AI-based simplification component. The AI component processes the medical content and generates an explanation in simple language. The simplified output is returned to the frontend and displayed on the webpage for the user.

4.3.7 Voice Explanation Module

The Voice Explanation Module converts the simplified medical information into spoken audio. It helps users who have difficulty reading or understanding lengthy medical reports.

1.Frontend Technologies: HTML5 provides the voice output interface, CSS3 styles the controls, and JavaScript manages speech-related functions.

2.Backend Technology: Python Flask provides the simplified text to the frontend. The actual speech playback is handled by the browser.

3.Tools Used: Web Speech API – Speech Synthesis.

4.Working Procedure:

After the simplified explanation is displayed, the user can select the voice output option. JavaScript passes the simplified text to the browser's Speech Synthesis engine. The selected language and an available matching voice are configured. The browser then converts the text into spoken audio, allowing users to listen to their medical report explanation.

4.3.8 Health Risk Indication Module

The Health Risk Indication Module provides a basic indication based on the information extracted from the medical report. It helps users identify report information that may require further attention.

1.Frontend Technologies: HTML5 structures the risk indication section, CSS3 styles the displayed information, and JavaScript presents the result.

2.Backend Technology: Python Flask coordinates the report information and the configured risk-indication logic.

3.Tools Used: Extracted medical report text and the application's configured processing logic.

4.Working Procedure:

After extracting and processing the medical report information, the application evaluates the available content using its configured indication logic. The result is displayed on the webpage as a basic health-risk indication. This feature is intended only for informational support and does not provide a medical diagnosis or replace professional medical advice.

4.3.9 Report History Module

The Report History Module allows users to access previously processed medical reports or their available summaries. It helps users refer to their earlier report information.

1.Frontend Technologies: HTML5 structures the history section, CSS3 provides styling, and JavaScript manages the display and retrieval of stored information.

2.Backend Technology: The current implementation uses browser local storage, so a separate database backend is not required for this history feature.

3.Tools Used: JavaScript Local Storage.

4.Working Procedure:

After processing a medical report, the application stores the available report history or summary in the browser's local storage. When the user opens the history section, JavaScript retrieves the stored information and displays the previous report entries. This allows users to access their earlier information through the same browser and device.

Important technical note: The exact AI model, database and authentication framework have not been confirmed in the available project details. Therefore, they are not mentioned as specific technologies. Also, browser voice availability depends on the device and supported languages.

4.4 EXPERIMENT ANALYSIS

OUTPUT:

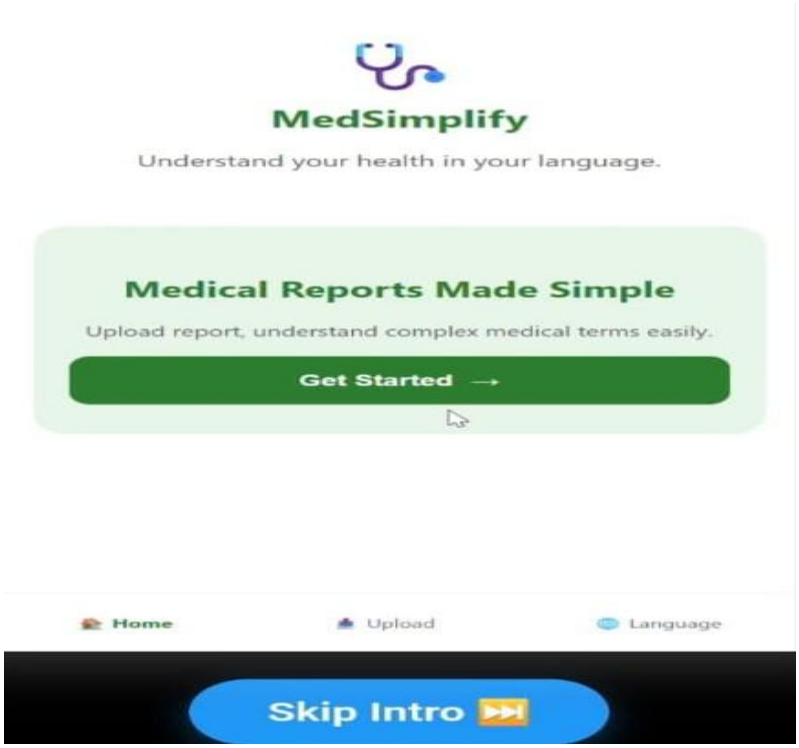


Figure 4.1 MRV Welcome Screen

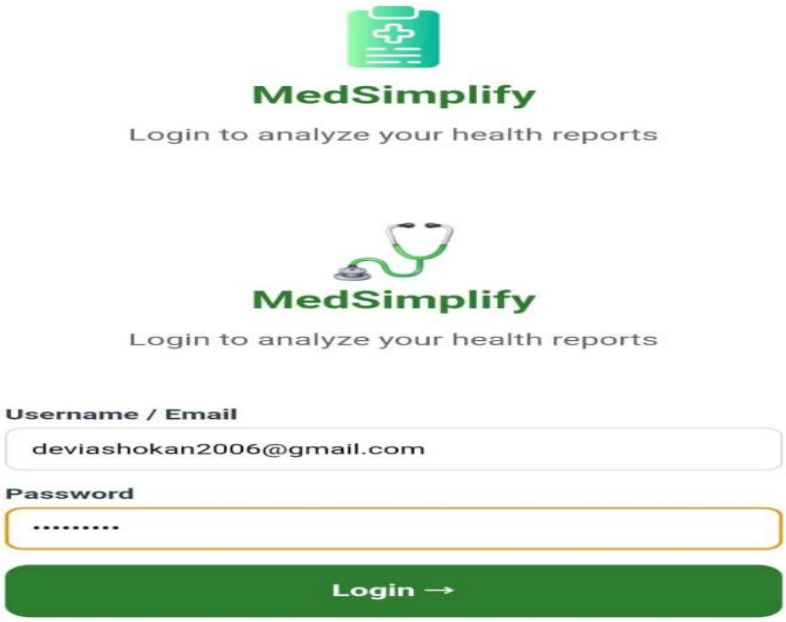


Figure 4.2 User Login Interface

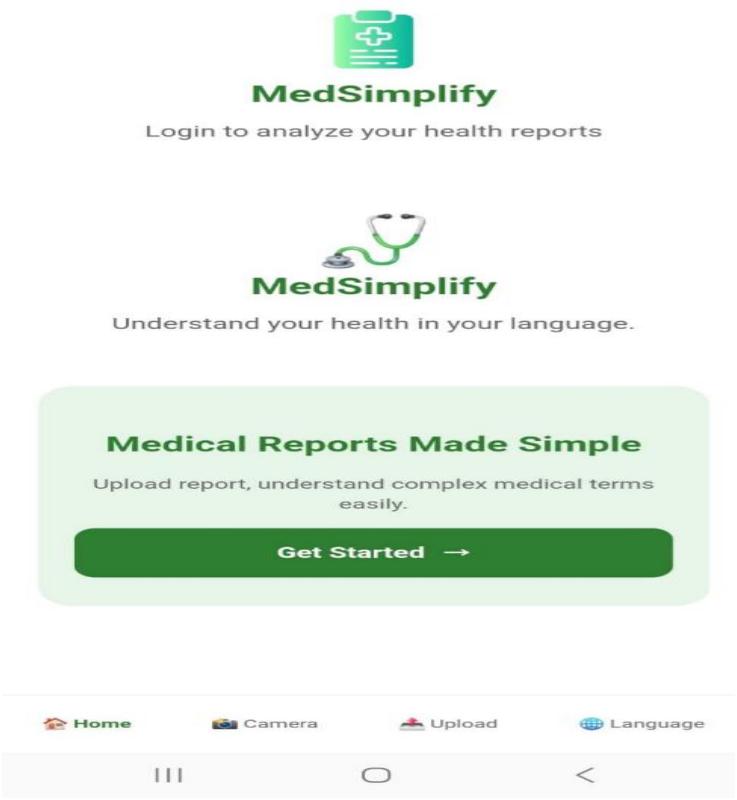


Figure 4.3 MRV Home Screen

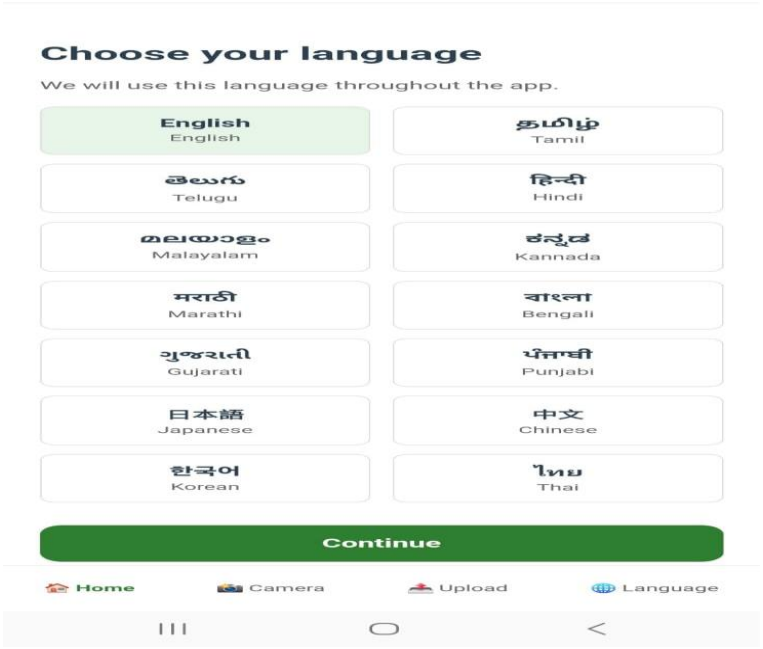


Figure 4.4 Language Selection Interface

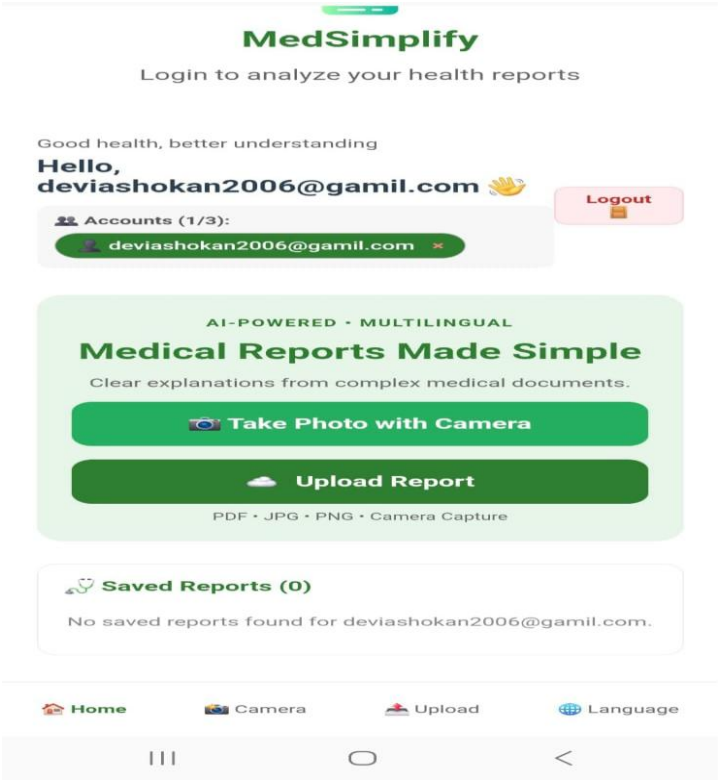


Figure 4.5 Medical Report Upload Interface

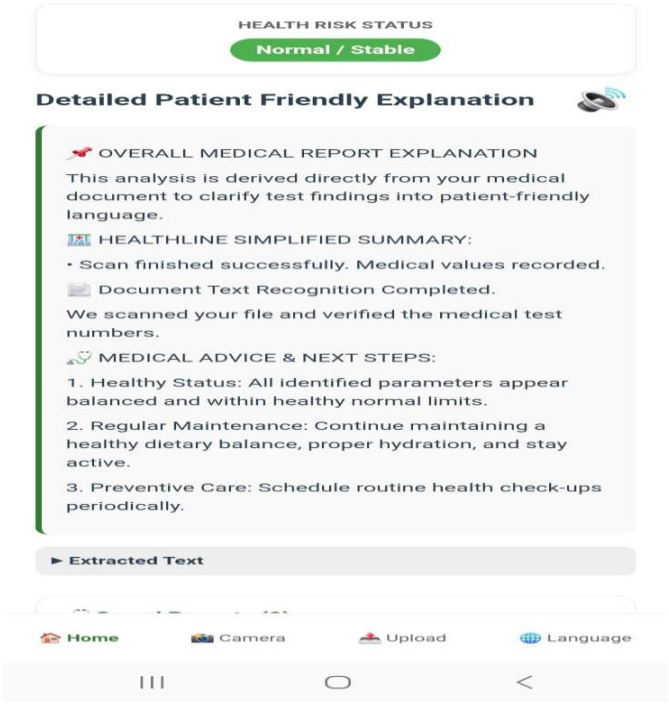


Figure 4.6 AI Powered Medical Report Simplification

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

The project titled “MRV: Interpreting Medical Reports into Voice in Regional Languages” aims to make medical information easier to understand by converting complex medical report content into simple language and voice explanations. Medical reports often contain technical terms that may be difficult for ordinary users to interpret. The proposed web-based system addresses this difficulty by combining document text extraction, Optical Character Recognition, AI-based text simplification and text-to-speech technology.

The system supports the upload of medical reports in PDF and image formats. PDF.js is used for extracting text from PDF documents, while Tesseract.js is used to recognize text from scanned report images. The extracted content is processed to generate simplified explanations in the user's selected language. Voice output further improves accessibility for users who may have difficulty reading or understanding written medical information.

The project also includes a basic health-risk indication and report-history facility to help users review important information and access previously processed reports. The multilingual interface is designed to support English, Tamil, Hindi, Telugu, Malayalam, Kannada, French and Japanese, subject to the availability and accuracy of the language-processing and browser voice components.

Overall, MRV provides a user-friendly approach to understanding medical reports and improves access to health information through regional-language explanations and voice assistance. The system is intended only as an information-support tool. Users should consult qualified healthcare professionals for medical interpretation, diagnosis and treatment decisions.

5.2 FUTURE WORK

1. Interactive Regional-Language Voice Assistant

In the future, MRV can include an interactive voice assistant that allows patients to ask questions about their medical reports in their preferred regional language. The system can respond with simple explanations and read the answers aloud using text-to-speech technology.

2. AI-Powered Report Comparison

In the future, MRV can use AI to compare a patient's current medical report with previous reports. The system can identify changes in medical test values and highlight whether each

value has increased, decreased, or remained stable. It can present these changes using simple summaries, tables, and visual indicators.

3.Medical Report Reminder Notifications

In the future, MRV can send notifications reminding patients to upload their latest medical reports after scheduled health check-ups. This will help users maintain an updated medical history and compare their results over time.

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SAMPLE SOURCE CODE

(index.html) – Main interface:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-
scale=1.0, user-scalable=no">
<title>MRV</title>
<!-- External Libraries for OCR & PDF Parsing -->
<script src="https://cdnjs.cloudflare.com/ajax/libs/pdf.js/3.11.174/pdf.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/tesseract.js@5/dist/tesseract.min.js"></script>


<style>
:root {
--primary: #2e7d32;
--primary-light: #e8f5e9;
--bg: #f4f7f6;
--card-bg: #ffffff;
--text: #2c3e50;
--subtext: #666666;
}


body {
font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
margin: 0;
background-color: var(--bg);
color: var(--text);
display: flex;
justify-content: center;
}


.app-container {
width: 100%;
max-width: 450px;
min-height: 100vh;
background: var(--card-bg);
box-shadow: 0 0 20px rgba(0,0,0,0.1);
position: relative;
padding-bottom: 70px;
box-sizing: border-box;
}


.screen { display: none; padding: 20px; }
```

```
.screen.active { display: block; }


/* INTRO VIDEO SCREEN OVERLAY - STRICT CENTER ALIGNMENT */
#screen-intro {
position: fixed !important;
top: 0 !important;
left: 0 !important;
width: 100vw !important;
height: 100vh !important;
background-color: #000000 !important;
z-index: 999999 !important;
display: flex !important;
flex-direction: column !important;
align-items: center !important;
justify-content: center !important;
padding: 0 !important;
margin: 0 !important;
}
.video-container {
width: 85%;
max-width: 300px;
display: flex;
flex-direction: column;
align-items: center;
justify-content: center;
gap: 15px;
}

video-container video {
width: 100%;
max-height: 45vh;
border-radius: 14px;
box-shadow: 0 8px 24px rgba(255, 255, 255, 0.2);
outline: none;
object-fit: contain;
 background-color: #000000;
}
#btnSkipIntro {
padding: 10px 24px;
font-size: 14px;
font-weight: 600;
background-color: #2196f3;
color: #ffffff;
border: none;
border-radius: 25px;
cursor: pointer;
box-shadow: 0 4px 12px rgba(33, 150, 243, 0.4);
transition: transform 0.2s;}
```

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MRV:Interpreting Medical Reports into Voice in Regional Languages

held during 21.10.2026 & 22.10.2026 organized by the Institute of CSE, SIMATS Engineering, SIMATS, Thandalam, Chennai. Your presence and engagement have contributed significantly to the discussions and knowledge-sharing on AI-driven Engineering Innovations.

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